



Girraween High School

2018 Year 12 Trial Higher School Certificate

Mathematics Extension 1

Time allowed: Two (2) Hours

(Plus 5 minutes reading time)

Instructions

- Attempt all questions.
- For Questions 1 -10, shade the circle for the letter corresponding to the correct answer on your answer sheet.
- For Questions 11 – 15, start each question on a new page. Each question should be clearly labelled.
- All necessary working must be shown for Questions 11– 15.
- Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used.
- A Mathematics reference sheet is provided.
- All diagrams are NOT TO SCALE.

Section 1

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Question 1

The point C which divides the interval between $A(14, -20)$ and $B(-4, 1)$ in the ratio 4: -1 is:

- (A) $(-10, 8)$ (B) $(20, -27)$ (C) $(2, -6)$ (D) $(8, -13)$

Question 2

$$\lim_{x \rightarrow 0} \frac{2x}{\sin 3x} =$$

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) 3

Question 3

The number of different ways of arranging the letters of the word MOORABOOL in a circle is

- (A) 362880 (B) 40 320 (C) 15 120 (D) 1680

Question 4

The remainder when $ax^3 + x^2 + x + 11$ is divided by $(2x + 3)$ is 5. The value of a is:

- (A) 1 (B) 2 (C) 3 (D) 4

Question 5

The co-efficient of x^2 in the expansion of $\left[2x + \frac{3}{x^2}\right]^{11}$ is

- (A) $\binom{11}{2} \times 2^9 \times 3^2$ (B) $\binom{11}{3} \times 2^8 \times 3^3$ (C) $\binom{11}{8} \times 2^3 \times 3^8$ (D) $\binom{11}{9} \times 2^2 \times 3^9$

Question 6

$$\cos 5x \cos 2x - \sin 5x \sin 2x =$$

- (A) $\cos 3x$ (B) $\sin 3x$ (C) $\cos 7x$ (D) $\sin 7x$

Examination continues on the following page

Question 7

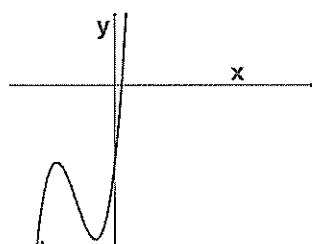
$$\int \sin^2 3x \cdot dx =$$

- (A) $\frac{1}{3} \sin^3 3x + C$ (B) $\frac{1}{2}x - \frac{1}{2} \cos 6x + C$ (C) $\frac{1}{2}x - \frac{1}{12} \cos 6x + C$
(D) $\frac{1}{2}x - \frac{1}{12} \sin 6x + C$

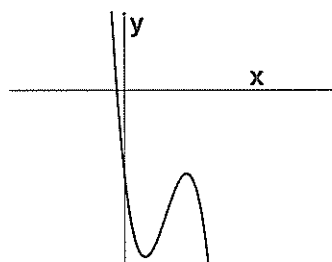
Question 8

Which of the graphs below is of $y = x^3 - 6x^2 + 9x - 4$?

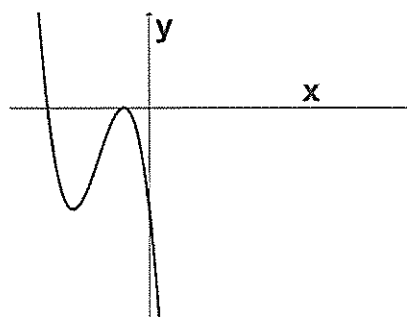
(A)



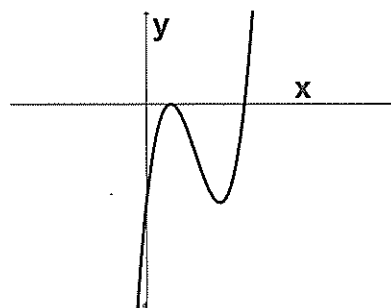
(B)



(C)



(D)



Question 9

The general solution of $\sin 2x = -\frac{1}{2}$ is

- (A) $x = n\pi + (-1)^n \times \frac{\pi}{6}$ (B) $x = \frac{n\pi}{2} + (-1)^n \times \frac{\pi}{12}$ (C) $x = n\pi + (-1)^{n+1} \times \frac{\pi}{6}$
(D) $x = \frac{n\pi}{2} + (-1)^{n+1} \times \frac{\pi}{12}$

Question 10

If $y = \sin^{-1}(5x)$, $\frac{dy}{dx} =$

- (A) $\frac{5}{\sqrt{1-25x^2}}$ (B) $\frac{1}{\sqrt{1-25x^2}}$ (C) $\frac{5}{\sqrt{25-x^2}}$ (D) $\frac{1}{\sqrt{25-x^2}}$

Examination continues on the following page

Section II

69 marks

Attempt Questions 11-15

Allow about 1 hour and 45 minutes for this section

Start all answers on a separate page in your answer booklet.

In Questions 11-15 your responses should include all relevant mathematical reasoning and/ or calculations.

Question 11 (16 Marks)

Marks

(a) Solve for x : $\frac{5}{3x-2} \geq -\frac{1}{4}$

3

(b) Find the acute angle between the lines $y = 2x - 1$ and $x + 3y = 2$

3

(c) Find the *exact* value of $\cos 75^\circ$.

2

(d) If α , β and γ are the roots of the polynomial equation

$2x^3 - x^2 + 3x - 4 = 0$, find the value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$

2

(e) Use the substitution $u = \cos x$ to find $\int_0^{\frac{\pi}{4}} \frac{\sin x}{1+\cos x} \cdot dx$

3

(f) Use the substitution $x = u^2$ to find $\int \frac{1}{1+\sqrt{x}} \cdot dx$

3

Question 12 (12 Marks)

(a) Use the method of mathematical induction to prove

$2 + 6 + 20 + \dots + (n^2 + n) = \frac{n}{3}(n+1)(n+2)$ for all positive integers n .

4

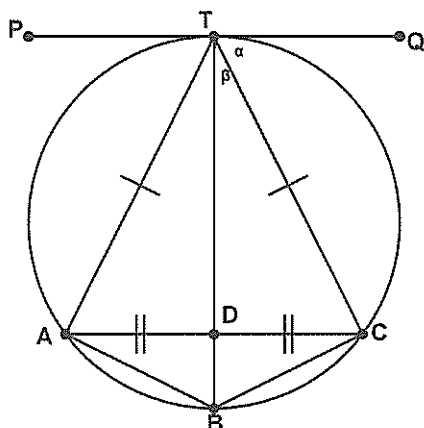
Question 12 continues on the following page

Question 12 (continued)**Marks**

(b) In the diagram below, line PTQ is a tangent to the circle $TABC$ at T .

BT bisects AC at D and $AT = CT$. $\angle QTC = \alpha$ and $\angle CTB = \beta$.

(see diagram)



(i) Copy the diagram in to your answer booklet and state why $\angle TAC = \alpha$.

1

(ii) Prove $\angle ATP = \alpha$.

2

(iii) Prove $\angle ATB = \beta$.

2

(iv) Prove TB is a diameter of the circle $TABC$.

3**Question 13 (14 Marks)****Marks**

(a) Sketch the graph of $y = 2\cos^{-1}\left(\frac{x}{3}\right)$, stating its domain and range.

3

(b) The probability that it will rain on a certain day in August in Sydney

is $\frac{2}{9}$. Assuming that the probability that it will rain on one day is *independent*

of the probability that it will rain on the next day:

(i) Find the probability that it will rain on 3 days in the next week.

2

(ii) Find the probability that it will rain on at least 2 days in the next week.

2

Question 13 continues on the following page

Question 13 (continued)**Marks**

(c) A particle is moving so that its position at time t is given by

$$x = 2\sin 3t + 2\sqrt{3}\cos 3t.$$

(i) By showing that $\ddot{x} = -n^2x$, n a real number, prove that the particle is moving with simple harmonic motion.

1

(ii) By expressing the particle's position at time t in the form $x = R\cos(3t - \alpha)$, find the period and amplitude of the motion.

3

(iii) Find the first time that the particle reaches $x = 0$ and find its velocity and acceleration then.

3**Question 14 (14 Marks)**

(a) A particle is moving with simple harmonic motion about $x = 0$ with acceleration $\ddot{x} = -n^2x$ and amplitude a . (Note that in this question a refers to the amplitude of the motion).

(i) Show that $v^2 = n^2(a^2 - x^2)$.

2

(ii) For this particle, $v = 10$ when $x = 5$ and $v = 5$ when $x = 7$. Find the period of the motion.

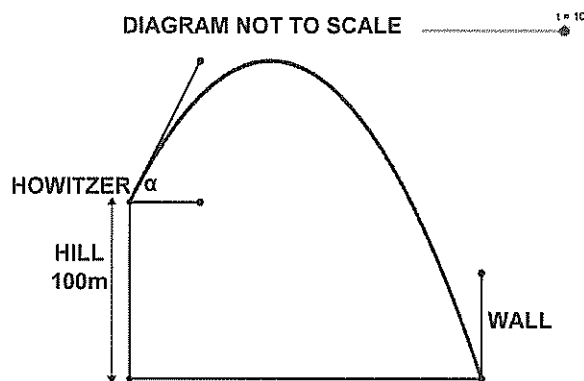
3

Question 14 continues on the following page

Question 14 (continued)**Marks**

(b) A howitzer with a muzzle velocity of 200m/s is located 100m up the side of a hill. It is firing shells at an angle of α to the horizontal.

(see diagram)



(i) Given that $\ddot{x} = 0$ and $\ddot{y} = -g$, show that $x = 200t\cos\alpha$ and $y = -\frac{1}{2}gt^2 + 200t\sin\alpha + 100$. 4

(ii) Show that $y = -\frac{gx^2}{80\,000}\sec^2\alpha + x\tan\alpha + 100$. 2

(iii) The shell is being aimed at the base of a wall 3000m away horizontally. 3
At which two angles must the shell be fired in order to hit this target?
(Let $g = 10\text{m/s}^2$).

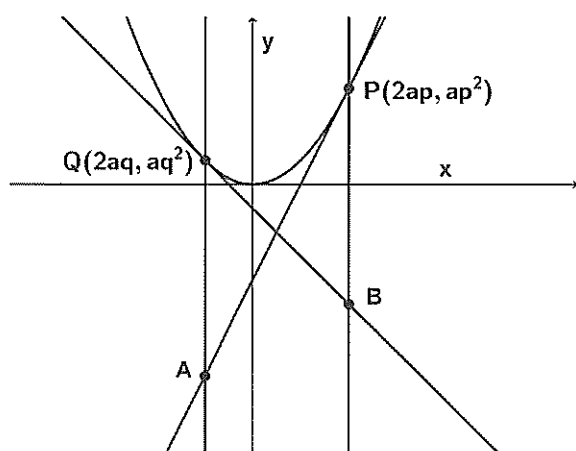
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Question 15 (13 Marks)**Marks**

(a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are variable points on the parabola $x^2 = 4ay$. A is the intersection of the tangent to the parabola at P and the line through Q parallel to the axis of the parabola, while B is the intersection of the tangent to the parabola at Q and the line through P parallel to the axis of the parabola. (see diagram)

The equation of the tangent to the parabola at P is $y = px - ap^2$

(Do NOT prove this!).



(i) Prove that $PQAB$ is a parallelogram.

3

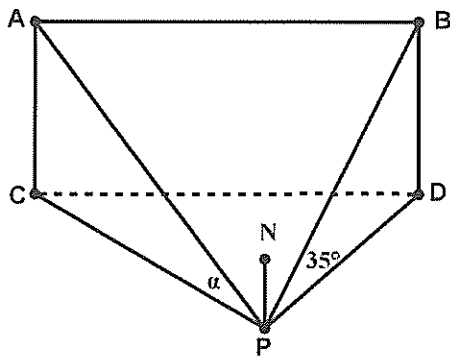
(ii) If $p > q$, find the area of the parallelogram in terms of p and q .

2

Question 15 continues on the following page

Question 15 (continued)**Marks**

(b) An aeroplane flying due East at 900km/h at a constant altitude is seen by an observer on the ground on a bearing of $290^\circ T$ at an angle of elevation of α . One minute later it is sighted by the observer at an angle of elevation of 35° and on a bearing of $050^\circ T$. If the aeroplane flies from A to B , the observer is at P and C and D are on the ground directly beneath A and B (see diagram)



- (i) Show that $AB = 15000\text{m}$ and $\angle CPD = 120^\circ$. 2
- (ii) Show that the height of the aeroplane above the ground is given by 3
- $$BD = \frac{15000 \sin 20^\circ \tan 35^\circ}{\sin 60^\circ}.$$
- (iii) Find the angle of elevation (α) when the aeroplane is first seen 3
- (answer to the nearest minute).

END OF EXAMINATION

Solutions

(1) A (2) B (3) D (4) B (5) B (6) C (7) D (8) D (9) D (10) A

$$Q.(1) \begin{array}{ccc} 14 & -4 & -20 \\ & \swarrow \searrow & \swarrow \searrow \\ 4 & -1 & 4 \end{array}$$

$$= \left(\frac{-14 - 16}{4 - 1}, \frac{20 + 4}{4 - 1} \right)$$

$$= (-10, 8) \quad (A)$$

$$Q.(2) = \frac{2}{3} \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} = \frac{2}{3} \times 1 = \frac{2}{3} \quad (B)$$

$$Q.(3) = \frac{8!}{4!}$$

= 1680 ways.

(D)

$$Q.(4) a \left(\frac{-3}{2} \right)^3 + \left(\frac{-3}{2} \right)^2 + \left(\frac{-3}{2} \right) + 11 = 5 \text{ by remainder theorem.}$$

$$\frac{-27a}{8} + \frac{47}{4}$$

$$= 5 \quad (B)$$

$$Q.(5) x^{11-k} \cdot x^{(x-2)^k} = x^2$$

$$11 - 3k = 2$$

$$\underline{k = 3}$$

$$\text{co-efficient} = T_4 \text{ co-eff}$$

$$= {}^{11}C_3 \times 2^8 \times 3^3 \quad (B)$$

$$Q.(6) = \cos(5x + 2x)$$

$$= \cos 7x \quad (C)$$

$$Q.(7) \int \sin^2 3x \, dx$$

$$= \frac{1}{2} \int (1 - \cos 6x) \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$$

$$= \frac{1}{2} x - \frac{1}{12} \sin 6x + C \quad (D)$$

$$Q.(8) y = x^3 - 6x^2 + 9x - 4$$

B & C out as $-x^3$

$$y' = 3x^2 - 12x + 9$$

$$y' = 0: x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1 \text{ or } 3 \rightarrow \text{must be } (D)$$

Multiple Choice Solutions (cont).

p. 2

Q. (9) $\sin 2x = -\frac{1}{2}$

$$2x = n\pi + (-1)^n \times -\frac{\pi}{6}$$

$$= n\pi + (-1)^{n+1} \times \frac{\pi}{6}$$

$$\therefore x = \frac{n\pi}{2} + (-1)^{n+1} \times \frac{\pi}{12} \quad \textcircled{D}$$

19 marks

Q. (10) $y = \sin^{-1}(5x)$

$$y' = \frac{1}{\sqrt{1-25x^2}} \times 5$$

$$= \frac{5}{\sqrt{1-25x^2}} \quad \textcircled{A}$$

Q. (11)(a) By critical points:

Equalities:

$$\frac{5}{3x-2} = -\frac{1}{4}$$

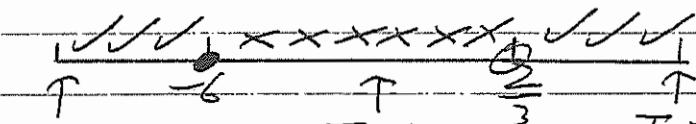
$$\times 4(3x-2)$$

$$20 = -3x + 2$$

$$\underline{-6 = x}$$

Discontinuity:

$$x \neq \frac{2}{3}$$



Test:
 $x = -7$

$$\frac{5}{3 \times -7 - 2} = \frac{5}{-23} \geq -\frac{1}{4} \checkmark$$

Test:
 $x = 0$

$$\frac{5}{3 \times 0 - 2} = -\frac{5}{2} \not\geq -\frac{1}{4} \times$$

Test:
 $x = 1$

$$\frac{5}{3 \times 1 - 2} = 5 \geq -\frac{1}{4} \checkmark$$

$$\therefore x \leq -6 \text{ \& } x > \frac{2}{3}$$

By "New York":

$$\frac{5}{3x-2} \geq -\frac{1}{4}$$

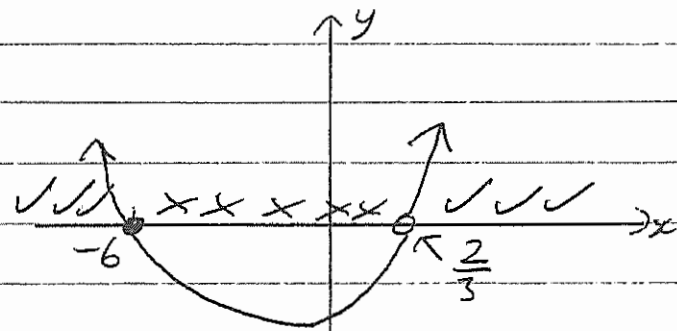
$$20(3x-2) \geq (3x-2)^2$$

$$60x - 40 \geq -9x^2 + 12x - 4$$

$$9x^2 + 48x - 36 \geq 0$$

$$3x^2 + 16x - 12 \geq 0$$

$$(3x-2)(x+6) \geq 0$$



$$x \leq -6 \text{ or } x \geq \frac{2}{3}$$

(b) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$= 2 + \frac{1}{3}$$

$$= \frac{7}{1 + 2 \times -\frac{1}{3}}$$

$$= 7$$

$$\underline{\theta = 81.52^\circ}$$

(c) $\cos 75^\circ$

$$= \cos(45^\circ + 30^\circ)$$

$$= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

Note: Equivalents such as $\frac{\sqrt{2}-\sqrt{3}}{2}$ are also OK.

Q (11) [continued]:

$$\begin{aligned}
 (d) \quad & \frac{1}{x\beta} + \frac{1}{x\gamma} + \frac{1}{\beta\gamma} \\
 &= \frac{x + \beta + \gamma}{x\beta\gamma} \\
 &= \frac{(\frac{1}{2})}{(\frac{4}{2})} \\
 &= \frac{1}{4}.
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad & \int_0^{\frac{\pi}{4}} \frac{\sin x}{1 + \cos x} dx \\
 &= - \int_0^{\frac{\pi}{4}} \frac{1}{1 + \cos x} \cdot (-\sin x) dx
 \end{aligned}$$

$$\text{Let } u = \cos x, du = -\sin x dx$$

$$\begin{aligned}
 &= - \int_1^{\frac{1}{\sqrt{2}}} \frac{1}{1+u} du \\
 &= - \left[\ln(1+u) \right]_1^{\frac{1}{\sqrt{2}}} \\
 &= \ln[1+u]_{\frac{1}{\sqrt{2}}} \\
 &= \ln 2 - \ln\left(1 + \frac{1}{\sqrt{2}}\right) \\
 &= 0.1583 \text{ [4DP]}.
 \end{aligned}$$

$ \begin{aligned} (f) \quad & \int \frac{1}{1+\sqrt{x}} dx \quad \text{Let } x = u^2, dx = 2u du \\ & \quad \quad \quad (u = \sqrt{x}). \\ &= \int \frac{1}{1+u} \cdot 2u du \\ &= \int \frac{2u+2}{1+u} - \frac{2}{1+u} du \\ &= \int 2 - \frac{2}{1+u} du \end{aligned} $	$ \begin{aligned} &= 2u - 2\ln(1+u) + C \\ &= 2\sqrt{x} - 2\ln(1+\sqrt{x}) + C \end{aligned} $
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p. 4

Q. (12)(a) Step 1: Show true for $n=1$.

LHS

$$= 2$$

RHS

$$= \frac{1}{3}(1+1)(1+2)$$

$$= 2$$

LHS = RHS. True for $n=1$.

Step 2: Assume true for $n=k$

$$\text{i.e. } 2+6+20+\dots+(k^2+k) = \frac{k}{3}(k+1)(k+2)$$

Step 3: Prove true for $n=k+1$

$$\text{i.e. } 2+6+20+\dots+(k^2+k)+(k+1)^2+(k+1) = \frac{(k+1)(k+2)(k+3)}{3}$$

LHS:

$$2+6+20+\dots+(k^2+k)+(k+1)^2+(k+1)$$
$$= \frac{k}{3}(k+1)(k+2) + (k+1)^2 + (k+1) \quad [\text{Using step 2 or by assumption}]$$

$$= \frac{(k+1)}{3} [k(k+2) + 3(k+1) + 3]$$

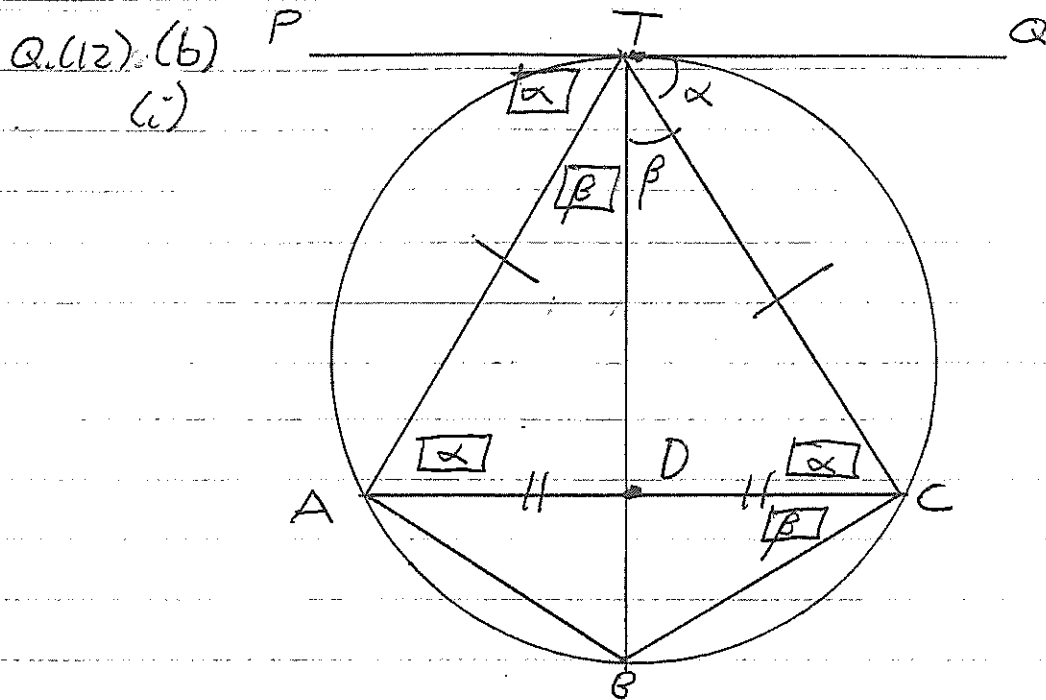
$$= \frac{(k+1)}{3} [k^2 + 5k + 6]$$

$$= \frac{(k+1)}{3} (k+2)(k+3)$$

$$= \text{RHS.}$$

∴ If it is true for $n=k$ it will be true for $n=k+1$.

∴ As it is true for $n=1$ it will be true for $n=1+1=2$ & so on for all positive integers n .



$\angle TAC = \alpha$ [$\angle$ between tangent & chord = $\angle$ in alternate segment].

(ii) $\angle ACT = \alpha$ [$\angle$ s opposite = sides in isosceles $\triangle TAC$].

$\angle ATP = \alpha$ [$\angle$ between tangent & chord = $\angle$ in alternate segment].

(iii) In $\triangle ATD$ & $\triangle CTD$

$AT = CT$ [data]

$AD = CD$ [data]

$\angle TAD = \angle TCD = \alpha$ [proven in (ii)]

$\triangle ATD \cong \triangle CTD$ [SAS].

$\therefore \angle ATD = \angle CTD = \beta$ [matching $\angle$ s, $\triangle ATD \cong \triangle CTD$].

(iv) $2\alpha + 2\beta = 180^\circ$ [$\angle$ s in straight line PTO].

$\therefore \alpha + \beta = 90^\circ$.

$\angle ACB = \angle ATB = \beta$ [$\angle$ s at circumference subtended by arc AB].

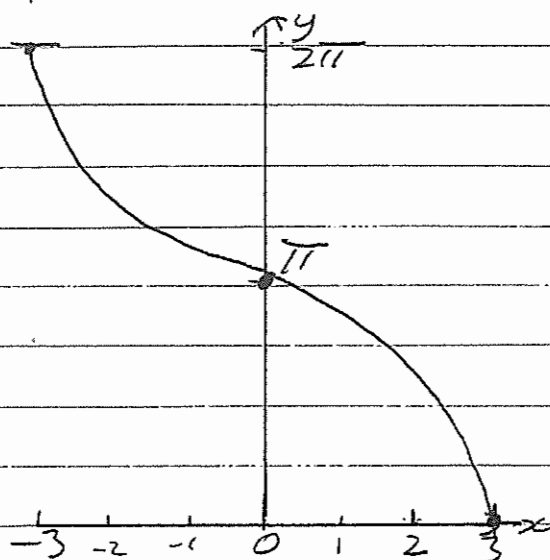
$\therefore \angle TCB = \alpha + \beta = 90^\circ$ [adjacent $\angle$ s].

TB is a diameter of circle $TACB$ [$\angle$ in semicircle = 90°].

Q. (13)(a) $y = 2\cos^{-1}\left(\frac{x}{3}\right)$

Key points:

$\frac{x}{3} = -1$	$\frac{x}{3} = 0$	$\frac{x}{3} = 1$	Domain: $-3 \leq x \leq 3$
$x = -3$	$x = 0$	$x = 3$	Range: $0 \leq y \leq 2\pi$
$y = 2\cos^{-1}(-1)$	$y = 2\cos^{-1}(0)$	$y = 2\cos^{-1}(1)$	
$= 2\pi$	$= \pi$	$= 0$	



(b)(i) $Pr = {}^7C_3 \times \left(\frac{2}{9}\right)^3 \times \left(\frac{7}{9}\right)^4$

$= 0.1405570473 \dots$

$\approx \underline{0.1406 \text{ (4DP)}}$

(ii) $Pr = 1 - [P(0 \text{ days}) + P(1 \text{ day})]$

$= 1 - \left[{}^7C_0 \times \left(\frac{2}{9}\right)^0 \times \left(\frac{7}{9}\right)^7 + {}^7C_1 \times \left(\frac{2}{9}\right)^1 \times \left(\frac{7}{9}\right)^6 \right]$

$= 0.4834528511 \dots$

$\approx \underline{0.4835 \text{ (4DP)}}$

(c)(i) $x = 2\sin 3t + 2\sqrt{3}\cos 3t$

$\dot{x} = 6\cos 3t - 6\sqrt{3}\sin 3t$

$\ddot{x} = -18\sin 3t - 18\sqrt{3}\cos 3t$

$= -9(2\sin 3t + 2\sqrt{3}\cos 3t)$

$\ddot{x} = -9x$

$\therefore$ Particle is moving with SHM.

p. 7

Q. (13)(c)(ii) Let $x = 2\sin 3t + 2\sqrt{3}\cos 3t \equiv R\cos(3t - \alpha)$

$\therefore 2\sin 3t + 2\sqrt{3}\cos 3t \equiv R\cos 3t \cos \alpha + R\sin 3t \sin \alpha$

Equating parts,

$$\begin{array}{l|l} 2\sin 3t = R\sin 3t \sin \alpha & 2\sqrt{3}\cos 3t = R\cos 3t \cos \alpha \\ 2 = R\sin \alpha \quad (1) & 2\sqrt{3} = R\cos \alpha \quad (2) \end{array}$$

Squaring & adding (1) & (2):

$$R^2(\sin^2 \alpha + \cos^2 \alpha) = 2^2 + (2\sqrt{3})^2$$

$$R = 4.$$

Sub. $R = 4$ in (1):

$$\frac{2}{4} = \sin \alpha$$

$$\frac{1}{2} = \sin \alpha$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore x = 4\cos(3t - \frac{\pi}{6})$$

Sub. $R = 4$ in (2):

$$\frac{2\sqrt{3}}{4} = \cos \alpha$$

$$\frac{\sqrt{3}}{2} = \cos \alpha$$

Period of motion = $\frac{2\pi}{3}$ seconds. Amplitude = 4m.

(iii) Particle reaches $x = 0$:

$$4\cos(3t - \frac{\pi}{6}) = 0$$

$$\cos(3t - \frac{\pi}{6}) = 0$$

$$3t - \frac{\pi}{6} = \frac{\pi}{2}$$

$$t = \frac{2\pi}{9} \text{ seconds.}$$

When $t = \frac{2\pi}{9}$ seconds,

$$\dot{x} = -12\sin(3t - \frac{\pi}{6})$$

$$= -12\sin(3 \times \frac{2\pi}{9} - \frac{\pi}{6})$$

$$= -12\sin \frac{\pi}{2}$$

$$= -12 \text{ m/s.}$$

$$\ddot{x} = 36\cos(3t - \frac{\pi}{6})$$

$$= 36\cos \frac{\pi}{2}$$

$$= 0 \text{ m/s}^2$$

[could also use

$$\ddot{x} = -9x$$

$$= -9 \times 0 = 0$$

$$= 0 \text{ m/s}^2]$$

p-8

(14)(a) $\ddot{x} = -n^2 x$.

(i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -n^2 x$.

$$\frac{d}{dx} (v^2) = -2n^2 x$$

$$v^2 = \int -2n^2 x \cdot dx$$

$$= -n^2 x^2 + C$$

As $v=0$ when $x=a$,

$$0^2 = -n^2 a^2 + C$$

$$n^2 a^2 = C$$

$$v^2 = n^2 a^2 - n^2 x^2$$

$$v^2 = n^2 (a^2 - x^2)$$

(ii) $v=10$ when $x=5$. $10^2 = n^2 (a^2 - 5^2)$

$$100 = n^2 a^2 - 25n^2 \quad (1)$$

$v=5$ when $x=7$. $5^2 = n^2 (a^2 - 7^2)$

$$25 = n^2 a^2 - 49n^2 \quad (2)$$

(1) - (2) $75 = 24n^2$

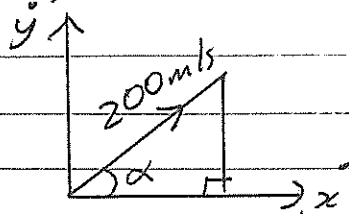
$$n = \frac{5\sqrt{2}}{4}$$

Period of motion = $\frac{2\pi}{n}$

$$= \frac{2\pi}{\left(\frac{5\sqrt{2}}{4}\right)}$$

$$= \frac{4\pi\sqrt{2}}{5} \text{ time units (seconds).}$$

Q. (14)(b)(i)



$$\text{Initial } \dot{x} = 200 \cos \alpha$$

$$\text{Initial } \dot{y} = 200 \sin \alpha$$

$$\ddot{x} = 0$$

$$\dot{x} = \int 0 \cdot dt$$

$$= C$$

$$= 200 \cos \alpha$$

$$x = \int 200 \cos \alpha \cdot dt$$

$$= 200t \cos \alpha + C$$

$$\text{As } x = 0 \text{ when } t = 0,$$

$$0 = 200 \times 0 \times \cos \alpha + C$$

$$0 = C$$

$$x = 200t \cos \alpha \quad (1)$$

$$\ddot{y} = -g$$

$$\dot{y} = \int -g \cdot dt$$

$$= -gt + C$$

$$\text{As } \dot{y} = 200 \sin \alpha \text{ when } t = 0$$

$$200 \sin \alpha = -g \times 0 + C$$

$$\dot{y} = -gt + 200 \sin \alpha$$

$$y = \int -gt + 200 \sin \alpha \cdot dt$$

$$= -\frac{1}{2}gt^2 + 200t \sin \alpha + C$$

$$\text{As } y = 100 \text{ when } t = 0$$

$$100 = -\frac{1}{2} \times g \times 0^2 + 200 \times 0 \times \sin \alpha + C$$

$$y = -\frac{1}{2}gt^2 + 200t \sin \alpha + 100 \quad (2)$$

$$(ii) x = 200t \cos \alpha$$

$$\frac{x}{200 \cos \alpha} = t \quad (3)$$

Sub. (3) in (2):

$$y = -\frac{1}{2} \times g \times \left(\frac{x}{200 \cos \alpha} \right)^2 + 200 \sin \alpha \times \frac{x}{200 \cos \alpha} + 100$$

$$y = \frac{-g x^2 \sec^2 \alpha}{80000} + x \tan \alpha + 100$$

$$(iii) y = 0 \text{ when } x = 3000 \text{ (} g = 10 \text{)}.$$

$$0 = \frac{-10 \times 3000^2}{80000} (1 + \tan^2 \alpha) + 3000 \tan \alpha + 100$$

$$0 = -1125(1 + \tan^2 \alpha) + 3000 \tan \alpha + 100 = -25$$

$$0 = 45 \tan^2 \alpha - 120 \tan \alpha + 41$$

$$\tan \alpha = \frac{120 \pm \sqrt{(-120)^2 - 4 \times 45 \times 41}}{2 \times 45}$$

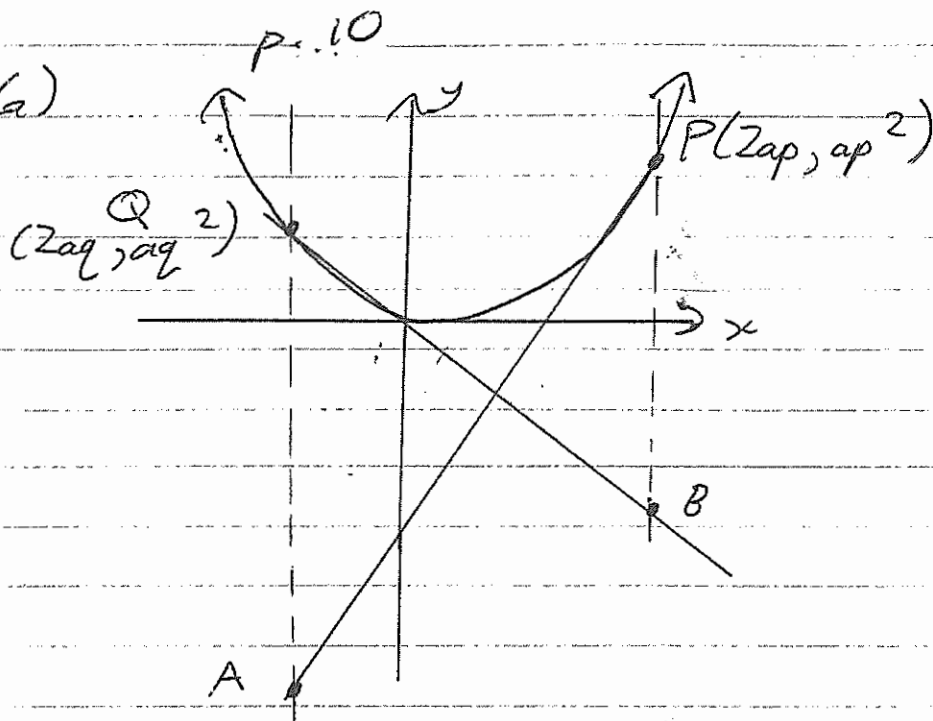
$$\tan \alpha = \frac{120 \pm \sqrt{7020}}{90}$$

$$\alpha = 66^\circ 10' \text{ or}$$

$$\alpha = 21^\circ 55'$$

Shell needs to be fired at $21^\circ 55'$ or $66^\circ 10'$ to hit target.

Q. (15) (a)



(i) Line QA is $x = 2aq$.

Finding y co-ordinate of A:

$$\begin{aligned} y &= px - ap^2 \\ &= p \times 2aq - ap^2 \\ &= 2apq - ap^2 \end{aligned}$$

$$\therefore A = (2aq, 2apq - ap^2)$$

$$\text{Similarly, } B = (2ap, 2apq - aq^2)$$

To prove parallelogram

Either

Or

$$\begin{aligned} \text{Distance QA} &= aq^2 - (2apq - ap^2) \\ &= a(q^2 - 2pq + p^2) \\ &= a(q - p)^2 \end{aligned}$$

$$\begin{aligned} m_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{a(p - q)(p + q)}{2a(p - q)} \\ &= \frac{p + q}{2} \end{aligned}$$

$$\begin{aligned} \text{Similarly, distance PB} &= ap^2 - (2apq - aq^2) \\ &= a(p^2 - 2pq + q^2) \\ &= a(p - q)^2 \end{aligned}$$

$$\begin{aligned} m_{BA} &= \frac{2apq - aq^2 - (2apq - ap^2)}{2ap - 2aq} \\ &= \frac{ap^2 - aq^2}{2a(p - q)} \\ &= \frac{a(p - q)(p + q)}{2a(p - q)} \\ &= \frac{p + q}{2} \end{aligned}$$

$\therefore$ PQAB is a parallelogram
[1 pair of opposite sides $\parallel$ & 11].

Note also that finding
midpoint PA & midpoint QB
& finding they coincide
[11 as diagonals bisect each
other].

or showing $PQ = BA$ as well
as $QA = PB$ [both pairs of opposite
sides $\cong$] are also acceptable.

$\therefore$ PQAB is a parallelogram
[both pairs of opposite sides $\parallel$].

Q. (15)(a) (continued):

$$(ii) A = Lh$$

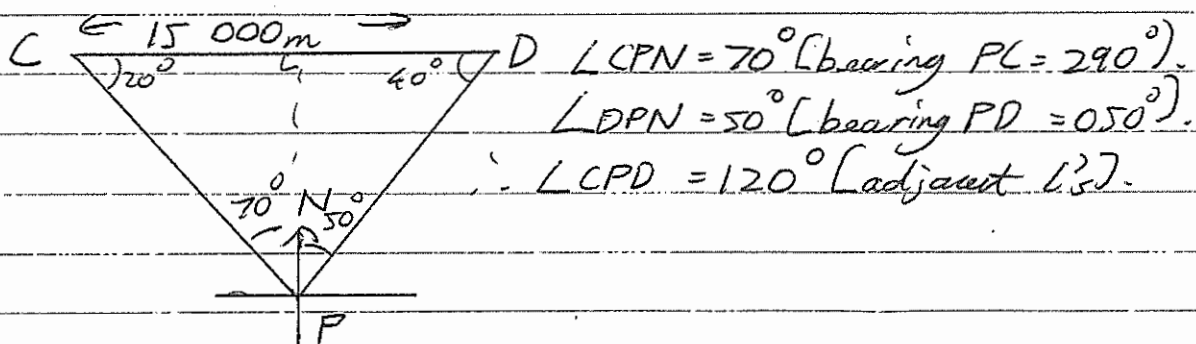
$$= a(q-p)^2 \times (2ap - 2aq)$$

$$= a(p-q)^2 \times 2a(p-q) \quad [\text{Noting that } (p-q)^2 = (q-p)^2]$$

$$= 2a^2(p-q)^3$$

(b)(i) In 1 minute,

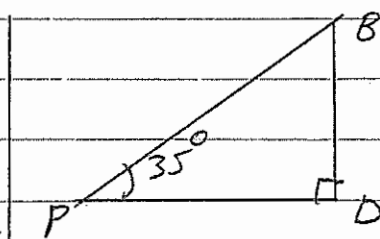
$$\text{plane flies } \frac{900\,000}{60} = 15\,000\text{m.}$$

(ii) Finding PD: Using $\triangle CPD$

$$\frac{PD}{\sin 20^\circ} = \frac{15000}{\sin 120^\circ}$$

$$\frac{PD}{\sin 20^\circ} = \frac{15000}{\sin 60^\circ} \quad [\text{as } \sin 60^\circ = \sin 120^\circ]$$

$$PD = \frac{15000 \sin 20^\circ}{\sin 60^\circ}$$



$$\tan 35^\circ = \frac{BD}{PD}$$

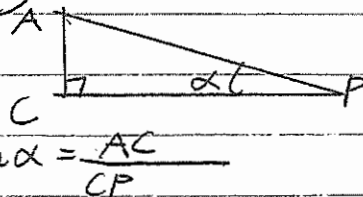
$$\therefore PD \tan 35^\circ = BD$$

$$\frac{15000 \sin 20^\circ \tan 35^\circ}{\sin 60^\circ} = BD = \text{Height.}$$

(iii) Using $\triangle CPD$

$$\frac{CP}{\sin 40^\circ} = \frac{15000}{\sin 60^\circ}$$

$$CP = \frac{15000 \sin 40^\circ}{\sin 60^\circ}$$

Using $\triangle ACP$ 

$$\tan \alpha = \frac{AC}{CP}$$

$$= \frac{15000 \sin 20^\circ \tan 35^\circ}{\sin 60^\circ} \times \frac{\sin 60^\circ}{15000 \sin 40^\circ}$$

$$\tan \alpha = \frac{\sin 20^\circ \tan 35^\circ}{\sin 40^\circ}$$

$$\alpha = 20.76^\circ$$

END OF
SOLUTIONS.